

二、CSCA 物理考试大纲 (2025 版)

1. 考试目的 (Purpose of the Exam)

本考试旨在评估来华留学生攻读中国本科理工科专业所需的物理核心素养，包括物理知识体系的构建、科学思维方法的掌握和跨学科应用能力的培养。这些核心素养不仅是大学物理课程的基础，更是机械、电子、材料等理工科专业学习的重要支撑，并为学生未来的科研创新和工程实践奠定坚实基础。

This exam aims to assess the core physics competencies required for international students to pursue undergraduate programs in science and engineering in China, including the establishment of a physics knowledge system, mastery of scientific thinking methods, and cultivation of interdisciplinary application competence. These core competencies are not only the foundation for university physics courses but also important support for studying science and engineering majors such as mechanical engineering, electronics, and materials science, laying a solid foundation for students' future scientific research innovation and engineering practice.

2. 考试形式与试卷结构 (Exam Format and Paper Structure)

- ◆ 考试时长: 60 分钟 (Exam Duration: 60 minutes)
- ◆ 满分: 100 分 (Full Score: 100 points)
- ◆ 语言: 中文 / 英文 (Language: Chinese/English)
- ◆ 题型: 单项选择题 (Question Type: Multiple-choice questions (single answer))
- ◆ 题量: 共 48 题 (Number of Questions: 48 in total)
- ◆ 内容模块 (Content Modules) :
 1. 力学 (Mechanics)
 2. 电磁学 (Electromagnetism)
 3. 热学 (Thermodynamics)
 4. 光学 (Optics)
 5. 近代物理 (Modern Physics)

3. 考试内容与范围 (Exam Content and Scope)

- ◆ 力学 (Mechanics)
 1. 运动学: 位移、速度、加速度, 匀变速直线运动, 自由落体运动 (Kinematics: Displacement, velocity, acceleration, uniformly accelerated linear motion, free fall motion)
 2. 牛顿运动定律及其应用 (Newton's Laws of Motion and their applications)
 3. 动量与冲量, 动量守恒定律 (Momentum and impulse, Law of Conservation of Momentum)
 4. 功与能, 机械能守恒定律 (Work and energy, Law of Conservation of Mechanical Energy)
 5. 圆周运动与万有引力 (Circular motion and Universal Gravitation)
 6. 简谐振动与机械波 (Simple Harmonic Motion and Mechanical Waves)
- ◆ 电磁学 (Electromagnetism)

1. 静电场：库仑定律，电场强度，电势 (Electrostatic Field: Coulomb's Law, electric field intensity, electric potential)
2. 直流电路：欧姆定律，串并联电路 (DC Circuits: Ohm's Law, series and parallel circuits)
3. 磁场：磁感应强度，安培力，洛伦兹力 (Magnetic Field: Magnetic Flux Density, Ampere's Force, Lorentz Force)
4. 电磁感应：法拉第定律，楞次定律 (Electromagnetic Induction: Faraday's Law, Lenz's Law)

◆ 热学 (Thermodynamics)

1. 分子动理论 (Kinetic Theory of Gases)
2. 理想气体状态方程 (Ideal Gas Law)
3. 热力学第一定律 (First Law of Thermodynamics)

◆ 光学 (Optics)

1. 几何光学：反射定律，折射定律 (Geometrical Optics: Law of Reflection, Law of Refraction)
2. 物理光学：干涉，衍射 (Physical Optics: Interference, Diffraction)

◆ 近代物理 (Modern Physics)

1. 光电效应 (Photoelectric Effect)
2. 原子结构 (Atomic Structure)
3. 核物理基础 (Fundamentals of Nuclear Physics)